N-F-00170d(Army-GL)
September 29, 1967
SUPERSEDING
Int. Fed. Spec. N-F-00170c(GL)
September 6, 1966

INTERIM FEDERAL SPECIFICATION

FEED, DRY, FOR DOGS

This interim Federal specification was developed by the U. S. Army Natick Laboratories, Natick, Mass., 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized Federal agencies to use this interim Federal specification as a valid exception to Federal Specification N-F-170b dated September 10, 1964.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for the preparation and packaging of a cooked, dry and complete feed capable of providing a suitable ration for dogs. The product may be held up to nine months prior to consumption.

1.2 Classification. The product covered by this specification shall be of the following compositions, forms of composition 1, and styles of composition 2, as specified (see 6.1):

Composition 1 - Dry

Form

A - Kibble

B - Pellets

C - Chunk (expanded)

D - Granules (expanded or crumbles)

Composition 2 - Semi-moist

Style

a. - Chunks

b. - Blocks

c. - Shreds

FSC 8710

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2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

L-C-110 - Cellophane (Coated and Uncoated Regenerated Cellulose Film.)

L-P-378 - Plastic film (Polyethylene Thin Gage).

TT-C-495 - Coatings, Exterior, for Tinned Food Cans.

UU-S-48 - Sacks, Shipping, Paper.

PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

PPP-B-636 - Box, Fiberboard.

PPP-C-29 - Canned Subsistence Items, Packaging and Packing of

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items.

MIL-L-10547 - Liners, Case, and Sheet Overwrap, Water-vaporproof or Waterproof, Flexible.

MIL-T-21330 - Treatment: Insect Resistant, for Paper.

MIL-L-35078 - Loads, Unit; Preparation of Nonperishable Subsistence In

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Governmental:

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder.

USDA-CMS-Visual Aid #1, Visual Aids for Inspection of Metal Containers.

(Copies may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Prices may be obtained from the Superintendent of Documents.)

U. S. Standards for Condition of Food Containers.

(Copies may be obtained from the Information Division, Consumer and Marketing Service, U. S. Department of Agriculture, Washington, D. C. 20250.)

Nongovernmental:

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Copies may be obtained from the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, D. C. 20044.)

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Methods of Vitamin Assay of the Association of Vitamin Chemists.

(Application for copies should be addressed to John Wiley and Sons, Inc., Interscience Publishers, 6057 Third Avenue, New York, N. Y. 10016.)

Feed Control - Official Publication

(Application for copies should be addressed to the Association of American Feed Control Officials, Inc., Office of the Treasurer, State Department of Agriculture - Room E-109, Charleston, West Virginia 25305.)

3. REQUIREMENTS

3.1 Materials. The product shall be prepared from the ingredients specified in table I. All ingredients shall be clean, sound, and free from mold, insect or insect part(s), filth, paint, glass, metal, or any other evidence of contamination or off condition. In addition, animal products, milk products, fats and oils and vitamin ingredients shall not be over 14 months old at start of processing. When used, frozen meat shall be uncured and shall not have been stored more than 180 days at a temperature of 0°F. or lower.

Table I. Ingredients.

Ingredient	NRC Reference Number 1/
Alfalfa products	
Dehydrated alfalfa meal	
20% crude protein	078-8000
17% crude protein	078-8000
Animal products	
Meat, beef	103-4000
Meat, Horsemeat	103-4000
Meat, by-products	103-5500
Meat and bone meal	103-7000
Meat meal, 55% crude protein	103-8500
Animal liver meal	105-0500
Animal liver and glandular meal	105-2000
Dried meat solubles	105-5000

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Table I. Ingredients. (cont'd)

Ingredient	NRC Reference Number 1/
Poultry parts	613-5500
Poultry by-product meal	613-7000
Dried whole egg	None
Barley products	
Barley	128-6000
Chemical preservatives	
BHA	201-6500
BHT	201-8000
Ethoxyquin	321-9500
Potassium sorbate	609-8000
Propyl gallate	615-9500
Sorbic acid	678-3500
Fats and oils	
Animal fat, preserved with BHA	107-0000
Vegetable fat or oil, preserved with BHA	785-6000
Poultry fat preserved with BHA	107-0000
Mono and di glycerides	None
Grain sorghums (milo, kaffir, etc.)	
Ground grain sorghum	709-5500
Cracked grain sorghum	709-4000
Grain sorghum mill feed	710-1500
Rolled grain sorghum	710-3000

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Table I. Ingredients. (cont'd)

Ingredient	NRC Reference Number 1/
Maize	
Ground corn	493-4000
Corn flour	494-0000
Toasted corn flakes	494-6000
Corn gluten meal, 42% protein	494-9000
Hominy feed	495-9500
Kibbled corn	497-0000
Corn germ meal (dry milled)	497-6000
Malto dextrin	468-2750
Corn starch	None
Dextrose	None
Marine products	
Fish meal, 60% minimum protein	348-3500
Fish liver and glandular meal	348-8000
Dried fish solubles	349-1000
Milk products	
Dried buttermilk, feed grade	218-4500
Dried skimmed milk, feed grade	218-9000
Dried whey	219-5000
Cheese rind	221-0000
Cheese meal	None

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Table I. Ingredients. (cont'd)

Ingredient	NRC Reference Number 1/
Mineral products	
Calcium carbonate	206-0000
Iodized salt	659-0000
Bone meal, steamed	106-1000
Cobalt carbonate	277-4000
Cobalt sulfate	277-8500
Copper carbonate	302-6000
Copper oxide	303-5000
Copper sulfate	303-6500
Dicalcium phosphate	206-6000
Iron carbonate	325-4000
Iron oxide	408-0500
Magnesium oxide	460-1000
Manganese sulfate	504-9500
Manganese oxide	505-5500
Potassium chloride	609-9500
Potassium iodate	610-1000
Potassium iodide	610-2500
Tricalcium phosphate	780-3500
Zinc oxide	851-3000
Zinc sulfate	851-4500
Sodium iron pyrophosphate	None
Titanium dioxide	None

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Table I. Ingredients. (cont'd)

Ingredient	NRC Reference Number 1/
Miscellaneous products	
Dried beet pulp	144-2000
Dried tomato pomace	780-0500
Cereal food fines	222-5000
Dried bakery product	114-2000
Oat products	
Oat groats	559-1000
Feeding oat meal	559-4000
Oat mill by-product	560-0000
Oat flour	None
Soybean products	
Soybean hulls	723-8000
Soy grits	724-1000
Soy flour	724-2500
Soybean meal	724-4000 724-5500
Soybean meal, dehulled, solvent extracted	724-7000
Special purpose products	
Aniseed	107-3000
Fennel	324-2000
Fenugreek seed	324-3500
Ginger	365-0000
Phosphoric acid	603-5000
Propylene glycol	616-1000
Onion powder	None

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Table I. Ingredients. (cont'd)

Ingredient	NRC Reference Number 1/
Garlic powder	None
Corn syrup	None
Sorbitol	None
Sucrose	None
Cellulose gum	None
Acacia (gum arabic)	None
Vegetable gums	None
Algin	None
Calcium citrate	None
Baking powder	None
Sodium carboxymethyl cellulose	None
Sodium cassinate	None
Certified food color	None
Carbon black	None
Vitamins	
Vitamin D_2 supplement	794-9000
Vitamin A AND D OIL	794-3000
D-Activated plant sterol	605-6000
Dried fermentation solubles	AAFCO 90.9 2/
Vitamin B_{12} supplement	794-4500
Vitamin E supplement	795-0500
Riboflavin supplement	629-6000
Vitamin A supplement	793-8500
Calcium pantothenate	205-4000

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Table I. Ingredients. (cont'd)

Ingredient	NRC Reference Number 1/
Choline chloride	226-7000
Choline citrate	None
Folic acid	356-6000
Menadione sodium bisulfite	508-7000
Niacin: Nicotinic acid	531-6500
Pyridoxine hydrochloride	617-0000
Thiamine: Thiamine hydrochloride	749-9000
Thiamine mononitrate	750-0500
Vitamin A palmitate	794-1500
Wheat products	
Wheat bran	819-3500
Wheat feed flour	819-5000
Wheat germ meal	819-6500
Wheat mill run	799-4000
Wheat middlings	819-8000
Wheat red dog	820-1000
Wheat starch	None
Yeast	
Irradiated dried yeast	847-1000
Brewers dried yeast	847-2500
Torula dried yeast	847-7000

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1/ The permanent identification number has been assigned to feed or nutrient by Animal Nutrition Sub-committee of National Research Council (NRC) and as listed in Feed Control.

2/ Reference No. listed in Feed Control.

3.2 Processing. The product shall be prepared from any combination of ingredients listed in table I necessary to yield a product which will comply with 3.3. For composition 1 product, not less than 18 percent, by weight of the total protein, shall be protein derived from animal products; for composition 2 product, not less than 18 percent of the total protein shall be derived from fresh meat or poultry, or meat, fish and poultry meals.

3.3 Finished product. The product shall be uniform in appearance as pertains to color and texture, as applicable, and shall be free from foreign materials (e.g. charred particles). Composition 1 product shall comply with the requirements of table II for piece size, and table III for chemical analysis. Composition 2, style b (blocks) shall weigh not less than 3 ounces. Composition 2, style a (chunks) shall comply with the requirements of table II for piece size. Both styles shall bomply with table III for chemical analysis.

3.3.1 Palatability shall be determined by feeding exclusively once a day at a rate of 35 calories per pound body weight to five dogs for five days. The dogs shall be working German Shepards that have been fed the ration at the indicated rate for at least fourteen days prior to the test. Failure to eat more than 60 percent of an individual feeding shall be considered a nonacceptance.

Table II. Sieve requirements for composition 1 product (percent by weight)(maximum)

	Requirements						
	Retained on US #8	Pass through US #8	Pass through US #10	Pass through US #40	Retained on US #3/8	Retained on US #5/8	Retained on US 1 inch
Form A, kibble	-	5	-	-	-	0	-
Form B, pellets	-	8	-	-	-	-	0
Form C, chunk, expanded	5	-	-	-	-	-	0
Form D, granules	-	-	15	-	5	-	-

Style a - Not less than 15 pieces per 6 ounce package.

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Table III. Analytical requirements.

Nutrients	Minimum value Composition 1	Minimum value Composition 2	Maximum value Composition 1	Maximum value Composition 2
	Percent by weight			
Crude protein	23.0	22.0	-	-
Crude fat	7.0	5.0	-	-
Crude fiber	-	-	5.0	3.5
Ash	-	-	10.0	10.0
Moisture	-	19.0	10.0	30.0
Calcium	1.0	1.0	-	-
Phosphorous	0.8	0.8	-	-
Potassium	0.8	0.8	-	-
Sodium chloride	-	-	1.4	1.4
Magnesium	0.04	0.04	-	-
	Mg/pound of feed - both compositions (minimum)			
Iron	24.0		-	-
Copper	2.6		-	-
Manganese	2.0		-	-
Iodine	0.6		-	-
Vitamin E	10.0		-	-
Vitamin B_{12}	0.01		-	-
Folic acid	0.07		-	-
Thiamine	0.5		-	-
Pyridoxime	0.5		-	-
Pantothenic acid	2.0		-	-

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Table III. Analytical requirements. (cont'd)

Nutrients	Minimum value Composition 1	Minimum value Composition 2	Maximum value Composition 1	Maximum value Composition 2
Niacin	5.0		-	-
Choline	400		-	-
		IU/pound of feed - both compositions		
Vitamin A	2000		-	-
Vitamin D	120		-	-

3.4 Federal Food, Drug and Cosmetic Act. All deliveries shall conform to the applicable provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.5 Workmanship.

3.5.1 For civil and military agencies. The product shall be processed, packaged and stored under modern sanitary conditions and in accordance with good commercial practice.

3.6 Storage of manufactured feed. The product shall not be stored more than the number of days indicated below:

(a) Level A packing - 30 days prior to shipment to first destination.

(b) Level C packing - 30 days prior to delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. Samples of the finished product covered by this specification and samples of any materials, components, or constituents entering into the preparation of the finished product may be taken any time by the Government inspector to determine compliance with the requirements.

4.1.1 Certificate of compliance (unless otherwise specified). Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.

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4.2.1 General. Inspection shall be made by the procuring agency or a duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing and packaging, at any suitable point in transit, and after delivery to the point of destination.

4.2.1.1 Condition inspection of containers. When inspection of product for quality and condition is performed by the U. S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply except for level A packing. (For applicable documents governing condition inspection of containers for military inspection agencies, including level A packing, see 4.2.6.3 and 5.2.1.1.

4.2.2 Factory inspection. The passing as satisfactory of any detail of processing or materials shall not relieve the supplier of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.2.3 Final inspection. Unless otherwise specified (see 6.1), final inspection shall be made after delivery to point of destination.

4.2.4 Responsibility for inspection. The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to perform the examination and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

4.2.4.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2.5 Preacceptance inspection (plant sanitation inspection.)

4.2.5.1 For Civil agencies. Inspection of plants producing the product shall be conducted upon request of the purchasing and contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval by the Military Veterinary Services.

4.2.5.1.1 For Military agencies. Award of contracts shall be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval by the Military Veterinary Services.

4.2.6 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated herein.

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4.2.6.1 Inspection of components (examination and testing). In accordance with 4.2.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification.

4.2.6.1.1 Inspection of ingredients. Conformance of ingredients to requirements contained in the publication "Feed Control" shall be ascertained by examination of pertinent labels, invoices or other valid documents. Dates on invoices shall be examined to determine the age of ingredients. In addition, each ingredient shall be examined organoleptically, as necessary, to determine conformance to condition requirements of 3.1. The sample unit for examination shall be approximately 1-pound of each major ingredient (comprises at least 1 percent of formulation) and 4 ounces of each minor ingredient, derived from a primary container. Five sample units shall be examined. In the event a lot consists of less than five containers, a sample unit shall be extracted from each container in the lot. Ingredients which do not conform to requirements of "Feed Control" (when applicable), or age requirement, or condition requirements, shall be unacceptable and use of such shall be cause for rejection of the lot or of the involved finished product.

4.2.6.1.2 Insect resistant treatment of sacks. Conformance to 5.2.1.1 as concerns insect resistant treatment of outer ply of sacks, shall be ascertained by examination of markings on sacks. If sacks are not marked "IRT" or the date of sack fabrication indicates the sack will be more than 75 days old at time of shipment of product, the sacks will be unacceptable and use of such shall be cause for rejection of the involved end item.

4.2.6.2 Process examination. Examination shall be made to determine compliance with the processing requirements of 3.2. Records of formulation shall be maintained on a percentage basis. In addition, (unless otherwise specified) the protein requirement for composition 1 or 2 product shall be ascertained by examination of certificate of compliance (unless otherwise specified). When a certificate of compliance is required to be verified (see 6.1) for the amount of animal protein in the product the following procedures shall be followed: A five pound composite sample derived from the number of containers indicated by inspection level S-2 shall be forwarded to the laboratory. Upon receipt of the results the following formula shall be used to determine compliance with the protein requirements: Test results times percentage of meat in formula equals the protein derived from animal products. Nonconformance to one or more of the requirements shall be cause for rejection of the lot or the involved quantity of finished product, as applicable.

4.2.6.3 Examination of end item. Examination of end item shall be in accordance with tables V through VIII and 4.2.6.3.1 through 4.2.6.3.4. Sampling plans and acceptance criteria for the aforementioned tables shall be expressed as defects per hundred units for table V, VI and VIII, and as percent defective for tables VII and IX.

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Table IV. Sampling plans and AQLs.

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major A	Major B	Minor	Total
V	I	1 filled and sealed primary container other than cans 1/	Primary containers	-	1.5	-	6.5
VI	I	1 filled and sealed can	Cans	0.25	1.5	4.0	-
VII	S-3	1 filled shipping container	Shipping containers	-	-	2.5	-
VIII	S-2	1 pound from a bag or sack; or 1 primary package from a case.	Primary containers	-	1.0	-	6.5
IX	S-3	1 filled and sealed can	Cans	1.0	-	-	-

1/ Applicable to all primary containers including level C shipping sack.

Table V. Examination of primary container (other than cans)

Category		Defect
Major B	Minor	
151		Open seam or seal; or tear or hole through container.
	201	Tear or hole through not more than 1 ply of sack.
	202	Objectionable odor.
	203	Not clean.
152		Nomenclature on shipping container missing, incomplete or illegible.
	204	Other required marking information on shipping container missing, incomplete or illegible.
153		Not type of packaging material specified.

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Table VI. External examination of cans 3/

Category			Defect
Major A	Major B	Minor	
101			False seam or improperly closed.
	151		Not type, size or style specified.
	152		Severe dent
	153		Dent causing sharp ridge
	155		Cable cut
	156		Pitted rust 1/
		201	Not clean 2/
		202	Moderate dent

1/ Rust that can be removed by wiping with a soft cloth will not be considered as pitted rust.

2/ Cans showing a very thin film of grease which is descernible to the touch but not readily descernable visually are considered to be clean.

3/ USDA-CMS-Visual Aid #1, (illustration of defects only) shall be used for classification of defects in lieu of current guidance referenced in PPP-C-29.

Table VII. Examination for net weight. 1/

Category	Defect
Minor	
201	Less than 49 pounds (Level A bag).
202	More than 2 percent under specified or marked net weight (sack or case).
203	Less than 161.9 grams (Level A package chunk or shred style).
204	Less than 77.1 grams (Level A package block style).

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1/ Report results to nearest ounce for level A bag, or level C sack or case. Report results to nearest 0.1 gram for "chunks" or "blocks". Lot shall be rejected if sample data indicate lot average net weight less than specified or marked net weight, as applicable. Average net weight of "chunks" shall be 170.4 grams and average net weight of "blocks" shall be 80.4 grams. In addition, style 2 product shall not be acceptable if marking or label indicates individual packages are less than 6 ounces net.

Table VIII. Examination of product. 1/

Category		Defect
Major B	Minor	
151		Moldy.
152		Burnt.
153		Not uniform in color or texture.
154		Odor not typical for the product.
	201	Less than 15 pieces (applicable to style A).

1/ Presence of foreign material (e.g., glass, metal, wood, stones, filth, insects, charred particles, etc.) shall be cause for rejection of the lot.

Table IX. Examination of leakage. 1/

Category	Defect
Major A	
101	Leakage.

1/ See 4.3.4

4.2.6.3.1 Examination of piece size (composition 1). Examination of piece size for composition 1 product shall be performed in accordance with 4.3.2. The sample for examination shall be a 5-pound composite of product derived from 5 primary containers. The sample shall be extracted from different areas of each container. Results shall be reported to the nearest whole percent. Nonconformance to piece size requirements shall be cause for rejection of the lot.

4.2.6.3.2 Examination for palatability. Examination shall be made to determine compliance with the palatability requirements of 3.3.1. Failure to meet palatability requirements shall be basis for rejection of the lot.

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4.2.6.3.3 Examination for storage of manufactured feed. Examination shall be made to determine compliance with the storage requirements of 3.6. Record of storage time shall be maintained. Nonconformance indicated by examination of records or date of pack shall be cause for rejection of the lot.

4.2.6.3.4 Examination of shipping sack (level A). The filled and closed shipping sacks shall be examined in accordance with the applicable criteria of UU-S-48. The following defects shall be included in the appropriate examination of UU-S-48 when empty plastic bags are required to be included in the shipping sacks: (1) Minor - Bags not secured as specified; (2) Major - Less than 10 bags: (3) Major - Size of bags inadequate to hold at least 2 pounds of product. In addition, the following marking defect shall be included in the appropriate examination: Major - Required markings on sack missing, incorrect or illegible. This defect shall be used in lieu of marking defects contained in UU-S-48, except those pertaining to sack manufacturer's compliance.

4.2.6.3.5 Examination of case liner. The filled and closed case liner shall be examined in accordance with criteria in the appendix of MIL-L-10547.

4.2.6.3.6 Examination of shipping case. The filled and closed shipping case shall be examined in accordance with the criteria in the appendix of PPP-B-636. In addition, the following defects shall be included in the examination : Major - marking missing, incorrect or illegible; minor - marking for level A or B pack indicates a net weight more than 27 pounds. For level C pack, only the major defect pertinent to marking shall apply.

4.2.6.3.7 Examination of palletized load. The palletized unit load shall be examined in accordance with the criteria in MIL-L-35078.

4.2.6.4 Testing of finished product. The finished product shall be tested for crude protein, crude fat, crude fiber, moisture, and ash in accordance with 4.3. The sample for testing shall be a one-pound composite derived from the number of containers indicated by inspection level S-2. Lot size shall be expressed in terms of containers. Results shall be reported to the nearest 0.1 percent. One or more test failures shall be cause for rejection of the lot. Sampling and rejection criteria shall be as stated above. Each test result shall be reported to the same decimal or unit as specified for the pertinent requirement.

4.2.6.5 Delivery. Product shall be rejected if shipment date or intended shipping date exceeds packing date (as indicated by container marking) by more than 30 days.

4.2.7 Standby test samples. The Government reserves the right to withdraw and hold standby test samples of components or finished product or both (quantity of which shall be not more than twice that required by this specification) for inspection purposes. These samples, if not used, will be returned to the supplier.

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4.3 Tests. Test procedures and controls which differ from those specified herein, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.3.1 Chemical analysis. Chemical analysis shall be made in accordance with the following methods in Official Methods of Analysis of the Association of Official Agricultural Chemists. In addition, analysis for Vitamins shall be made in accordance with Methods of Vitamin Assay of the Association of Vitamin Chemists:

Test	Method
Moisture	Moisture
Crude Protein	Crude Protein
Fat	Direct (ether extract)
Crude Fat (baked or expanded product)	Expanded Pet Food (acid hydrolysis)
Fiber	Crude Fiber
Ash	Ash
Total Protein	Kjeldahl
Mineral 1/	Method
Calcium	Atomic Absorption Spectrophotometry
Magnesium	"
Iron	"
Copper	"
Manganese	"
Iodine	Iodine and Mineral Mixed Feed
Phosphorus	Phosphorus in Feed
Potassium	Flame Photometric

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Test	Method
Sodium Chloride	Chloride and ash as Sodium Chlorides
Vitamins 1/	**Method**
Vitamin A	Colorimetric
Carotene	Chromatographic
Thiamine	Thiochrome
Riboflavin	Fluorometric
Choline	Reineckate
Vitamin D	Chemical
Vitamin E	Chemical
Niacin	Microbiological
Pantothenic Acid	"
Pyridoxine	"
Folic acid	"
Vitamin B_{12}	"

1/ Test shall be conducted only when specified (see 6.1)

4.3.2 Sieving test for finished product (complsition 1). Procedure pertinent to determine piece size of the various forms of composition 1 product shall be as follows:

(1) Nest the required sieves with the larger mesh sieve on top, and attach a collecting pan to the bottom sieve.

(2) Place a quantity of product equal to approximately 1/4 the volume of a sieve in the top sieve and attach a cover to the top sieve.

(3) Tip the sieves back and forth through a 90-degree arc and occasionally agitate the product by briskly moving sieves in a horizontal plane with a linear or circular motion. Sustain this procedure for three minutes.

(4) Follow the above procedures (1, 2, 3) until the five-pound sample is utilized.

(5) Weigh product retained on top sieve and passing through bottom sieve, respectively, and calculate percentage of each.

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4.3.3 Tin coating weights. Tin coating weights shall be tested in accordance with any method specified in PPP-C-96.

4.3.4 Leakage test. The filled and closed can shall be tested for leakage by submerging the can in water in such a manner that the can is lying on a side with not less than 3 inches of water over any portion of the can, and not less than 1-inch of water between the lower portion of the can and the bottom of the tank. The can shall be held in this position for not less than 30 seconds and shall then be rotated so that observation for leaks can be made with each of the four sides facing the top of the tank. The temperature of the water shall be 100°F. ± 5°F. A leak consists of a steady progression of bubbles. Isolated bubbles caused by entrapped air shall not be considered as a sign of leaks.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with 5.1.1 or 5.1.2, as specified (see 6.1).

5.1.1 Level A. The product shall be packaged in accordance with 5.1.1.1 or 5.1.1.2, as applicable.

5.1.1.1 Composition 1 product. Fifty pounds of the composition 1 product shall be packaged into a heat sealed plastic bag constructed of 0.005-inch thick polyethylene in accordance with type I of L-P-378.

5.1.1.2 Composition 2 product. Six ounces of chunks or shreds shall be packaged into a heat sealed bag, or a 3-ounce block shall be packaged into a conformi heat sealed wrap or bag. Bag and wrapping materials shall be 0.0011-inch thick cellophane complying with type VI or type VII, class 2, of L-C-110.

5.1.2 Level C. The product shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. The product shall be packed in accordance with 5.2.1, 5.2.2, or 5.2.3 as specified (see 6.1).

5.2.1 Level A. The product shall be packaged in accordance with 5.2.1.1 or 5.2.1.2, as applicable.

5.2.1.1 Composition 1 product. A fifty pound bag of composition 1 product, packaged as specified in 5.1.1.1, shall be packed into a multiwall paper shipping sack constructed in conformance with type I, class 2, or type II, sack number 4 or 4X, MB 1, level A pack, of UU-S-48. The outer surface of the sack shall be insect resistant treated in accordance with MIL-T-21330 and marked in accordance with UU-S-48. The top and bottom ends of the type II sack shall be sewn in accordance with UU-S-48 except that the stitching shall be reinforced with kraft paper tape that will prevent the thread from cutting into the wall of the sack.

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5.2.1.1.1 Individual ration bags. When specified (see 6.1), 10 empty plastic bags, constructed of 0.001-inch thick polyethylene in accordance with type I of L-P-378 and heat sealed on 3 sides, shall be placed on top of the filled and sealed polyethylene bag of product prior to closing the shipping sack.

5.2.1.2 Composition 2 product. Seventy-six 6 ounce bags of shreds or chunks or 144 wrapped blocks shall be packed in accordance with 5.2.1.2.1 or 5.2.1.2.2, as specified (see 6.1, 6.2 and 6.3).

5.2.1.2.1 Cans (see 6.2). The product shall be packed into a plastic bag constructed of 0.005-inch thick polyethylene complying with L-P-378. The bag shall be made from flat or roll stock or tubing, and shall be made to fit snugly into a 5-gallon square can. After filling the bag shall be heat sealed. The filled and sealed bag shall be placed into a 5-gallon square, metal can having approximate dimensions of 9-3/8-inches by 9-3/8-inches by 13-7/8-inches. The can shall be made throughout from not less than commercial 0.25-pound electrolytic tin plate (see 4.3.3) and shall have soldered side seams and double-seamed soldered or squeezed-on soldered top and bottom. The can shall be coated overall with a coating conforming to type I of TT-C-495. The can shall not leak when tested in accordance with 4.3.4.

5.2.1.2.1.1 The filled and sealed can shall be packed into a snug fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC, $V2_s$, of PPP-B-636.

5.2.1.2.2 Fiberboard box (see 6.3). The product, in a 0.005-inch thick polyethylene bag specified in 5.2.1.2.1 shall be packed into a snug-fitting fiberboard box with a liner, constructed, closed, and reinforced in accordance with style RSC-L, $V2_s$, of PPP-B-636. The liner shall be of the same material as the fiberboard box.

5.2.2 Level B, composition 2 product only. Composition 2 product, packed in a 0.005-inch thick polyethylene bag as specified in 5.2.1.2.1, shall be packed into a snug-fitting fiberboard box with a liner constructed and closed in accordance with style RSC-L, type CF or SF, class domestic, method II closure, of PPP-B-636. The liner shall be of the same material as the fiberboard box.

5.2.3 Level C. The product shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the rules or regulations applicable to the mode of transportation.

5.3 Unit loads. The product shall be assembled in unit loads in accordance with MIL-L-35078.

5.4 Labeling and marking.

5.4.1 Civil agencies.

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5.4.1.1 Labeling, polyethylene bags. Any labeling that complies with the Federal, Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder, and which shows crude protein, fat, fiber, ash and moisture content, caloric value per ounce, and instructions for feeding is acceptable. Alternatively, labeling information for composition 1 product shall be placed on the exterior of the multiwall shipping sack.

5.4.1.2 Shipping container. Shipping container shall be marked in accordance with Fed. Std. No. 123 and with such other information specified by the procuring agency (see 6.1).

5.4.2 Military agencies.

5.4.2.1 Labeling

5.4.2.1.1 Polyethylene bags. Labeling specified in 5.4.1.1 will be acceptable.

5.4.2.1.2 Metal cans. Metal cans shall be labeled in accordance with MIL-L-1497 and shall include the information specified in 5.4.1.1.

5.4.3 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.4 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078 and MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Composition, form (composition 1), and style (composition 2) of product required (see 1.2).

(c) When final inspection is to be made other than after delivery to the initial point of domestic destination (see 4.2.3).

(d) Levels of packaging and packing required (see 5.1 and 5.2).

(e) When individual ration bags are required (see 5.2.1.1.1).

(f) Additional labeling and marking required (see 5.4.1.1 and 5.4.1.2).

(g) When the certificate of compliance is required to be verified.

6.2 Packing specified in 5.2.1.2.1 is intended for shipment to and storage in tropical areas.

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6.3 Packing specified in 5.2.1.2.2 is intended for shipment to and storage in nontropical areas.

6.4 Destination inspection (For civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

Custodians:

Army - GL

Preparing activity:

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